

Introduction. Number systems.

Codes. Errors

1. Number systems:

Def. A number system is a given set of rules for representing and encoding information using a set of symbols called digits.

Classification:

1. Non-positional: The value/weight of a digit does not depend on its position within the number (Roman m.).
2. Positional: Each digit has an associated weight based on a base b .

Obs. A number N expressed in base b is written:

$$(N)_b = C_{m-1} \cdot b^{m-1} + C_{m-2} \cdot b^{m-2} + \dots + C_0 b^0 + C_{-1} b^{-1} + \dots + C_{-p} b^{-p}$$

b - base of system ($b \geq 1$)

C_i - digits, $0 \leq C_i \leq b-1$

m - number of digits in the integer part

p - number of digits in the fractional part

Obs. The radix point separates the integer part from the fractional part.

Def. In a given base b , the complement of a digit c is defined as:

$$\bar{c} = (b-1) - c$$

Common number systems:

1. Binary system ($b=2$)

$$\text{ex. } (1011.101)_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} = 11.625$$

2. Octal system ($b=8$)

- Each octal digit can be encoded using 3 bits

3. Hexadecimal ($b=16$)

- |————— // —————| 4 bits

Conversion between bases:

Case A: $b_1 < b_2$ (smaller to bigger base)

→ the number $(N)_{b_1}$ is expressed as a polynomial using powers of b_1 and evaluated using b_2 arithmetic.

Case B: $b_2 > b_1$ (bigger to smaller base)

→ Calculations are made using b_1 arithmetic:

- Integer part: Repeatedly divide by base b_2 . Collect the remainders from right to left until the quotient becomes 0

- Fractional part: Repeatedly multiply by b_2 . Collecting integer parts from left to right until the desired accuracy is reached.

ex. $b_1 = 10$ $N = 365.6875$

$b_2 = 2$

365		2
...		...
...		...
...		...

0		6875
1		3750
0		7500
1		5000
1		0000

=> fractional (N)
= 1011

2. Binary codes for decimal numbers:

Def. Weighted codes are ones in which every bit b_i has an associated weight w_i . The decimal digit value is expressed as:

$$N = b_3 w_3 + b_2 w_2 + b_1 w_1 + b_0 w_0$$

ex. BCD 8421, Aiken 2421, 632-3 code

Obso. 1) A num is self-complementary if the sum of the weights is 9.

2) For any decimal digit A, the code word for its complement $\bar{A} = 9 - A$ is obtained by inverting all bits.

Non-weighted codes:

1. Excess 3 code:

- Obtained by adding the binary value 3

to each BCD 421 code word.

- It is a non-weighted self-complementary code.

2. Gray code

- A reflected and cyclic code
- Any two consecutive code words differ by one single bit.
- An n -bit gray code is constructed by reflecting the $(n-1)$ bit code and appending a leading bit

3. Error detection

Parity bit method:

Consists of appending an extra bit to the leftmost position of the code word:

- Even parity:

- The parity bit is set to 0 if the nr. of 1s in the code is even

- The parity bit is set to 1 if the nr. of 1s is odd.

- Odd parity:

- The parity bit is set to 0 if the nr. of 1s is odd

- The parity bit is set to 1 if the nr. of 1s is even.

Rule: For successful error detection, both the source and destination must use the same parity type.

Exercises:

1. $(1001101)_4 \rightarrow$

$$01000001010001 = (1000001010001)_2$$

$$(001)(000)(001)(010)(001) = (10121)_8$$

$$(00011000101010001) = (1051)_{16}$$

2. $X = (04B45)_{16}$

$$0 = 0000$$

$$A = 1010$$

$$B = 1011$$

$$4 = 0100$$

$$5 = 0101$$

$$\Rightarrow X = (00001010101101000101)_2$$

$$X = (002223111)_4$$

$$X = (0125505)_8$$

3.

0	00	000
1	01	001
	11	011
	10	010
		110
		111
		101
		100

$$11 \rightarrow 110$$

$$12 \rightarrow 111$$

$$13 \rightarrow 101$$

$$14 \rightarrow 100$$

$$15 \rightarrow 0000$$

$$4. \quad 273.28125 = 5$$

$$\Rightarrow 1(273.28125) = 100010001$$

0	28125
1	56250
0	112500
0	225000
0	450000
1	900000

273	2
136	1
68	0
34	0
17	0
8	1
4	0
2	0
1	0
0	1

$$\Rightarrow 273.28125 = (100010001.01001)_2$$

$$= (0100010001.010010)_2 = (10101.102)_4$$

$$= (421.22)_8 = (111.42)_{16}$$

